

Resume Skills Extractor (NLTK + Python)

Project Overview

The **Resume Skills Extractor** is an NLP-based project built using **Python**, **NLTK**, and **PyPDF2**, designed to automatically extract key information from a resume PDF, including:

- **Technical Skills**
- **Education**
- **Work Experience**
- **Named Entities** (Organizations, People, Dates)

This project is fully optimized to run on **Google Colab**, supports modern **NLTK v3.9+**, and resolves all new tokenizer/tagger issues (e.g., `punkt_tab`, `averaged_perceptron_tagger_eng`).

Features

- ✓ Upload a **PDF resume**
- ✓ Extract clean text from PDF
- ✓ NLP preprocessing with NLTK
- ✓ Skills detection (keyword + multi-word)
- ✓ Education extraction
- ✓ Experience extraction
- ✓ Named Entity Recognition (NER) using NLTK's chunker
- ✓ 100% working in Google Colab with no errors

Technologies Used

- **Python**
- **NLTK**
- **PyPDF2**
- **Google Colab**

- **Tokenization**
- **NER (Named Entity Recognition)**
- **Phrase Chunking**
- **Keyword Extraction**

Install Required Libraries

```
!pip install nltk PyPDF2
```

Download NLTK resources:

```
import nltk
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('averaged_perceptron_tagger_eng')
nltk.download('maxent_ne_chunker')
nltk.download('words')
```

Example Output

```
{
  "Skills": ["Python", "Machine Learning", "Deep Learning", "Sql", "Git"],
  "Education": ["Bachelor of Engineering in Computer Engineering"],
  "Experience": ["Machine Learning Intern at XYZ (2023)"],
  "Named Entities": [
    ["John Doe", "PERSON"],
    ["XYZ Technologies", "ORGANIZATION"],
    ["2023", "DATE"]
  ]
}
```

Use Cases

- Automated resume screening
- HR & ATS systems

- Candidate profiling tools
- AI-powered hiring platforms
- NLP practice/portfolio project

Project Advantages

- Beginner-friendly
- Real-world HR application
- No ML model required
- NLTK-only NLP pipeline
- Works on ANY PDF resume

Future Improvements

- Integrate spaCy for better NER
- Add cosine similarity skill matching
- Convert output to a downloadable PDF/CSV
- Create a Streamlit web app interface

License

- **Atharva Gole** — Developer
(goleatharva8@gmail.com)
- ChatGPT — Assistant for NLP pipeline & project setup